What gridmicrotex draws

Each row is a LaTeX input and what the package makes of it, three to a line where they fit: accents, delimiters, operators, relations and symbols first, then further parts of LaTeX – units, equations, tables, diagrams, TikZ pictures, boxes and columns. A row is a test: it must be drawn with no warning. Where LaTeX has nothing to compare with, only the name is listed.

This PDF is rendered with this package.

Accents

`a'`	a'	`\tilde{a}`	$\tilde{a}$	`\mathring{g}`	$\mathring{g}$
`a''`	a''	`\widetilde{ac}`	$\widetilde{ac}$	`a^{\prime}`	$a^{\prime}$
`\utilde{AB}`	$\underset{\sim}{AB}$	`\acute{a}`	$\acute{a}$	`\vec{F}`	$\vec{F}$
`\bar{y}`	$\bar{y}$	`\breve{a}`	$\breve{a}$	`\check{a}`	$\check{a}$
`\dot{a}`	$\dot{a}$	`\overbrace{AB}`	$\overbrace{AB}$	`\ddot{a}`	$\ddot{a}$
`\underbrace{AB}`	$\underbrace{AB}$	`\dddot{a}`	$\dddot{a}$	`\overline{AB}`	$\overline{AB}$
`\overbracket{AB}`	$\overbracket{AB}$	`\ddddot{a}`	$\ddddot{a}$	`\underline{AB}`	$\underline{AB}$
`\underbracket{AB}`	$\underbracket{AB}$	`\grave{a}`	$\grave{a}$	`\underbar{X}`	$\underline{X}$
`\hat{\theta}`	$\hat{\theta}$	`\widehat{ac}`	$\widehat{ac}$	`\text{\'{a}}`	$\text{\'{a}}$
`\text{\~{a}}`	$\text{\~{a}}$	`\text{\.{a}}`	$\text{\.{a}}$	`\text{\`{a}}`	$\text{\`{a}}$
`\text{\={a}}`	$\text{\={a}}$	`\text{\"{a}}`	$\text{\"{a}}$	`\text{\v{a}}`	$\text{\v{a}}$
`\text{\^{a}}`	$\text{\^{a}}$	`\text{\u{a}}`	$\text{\u{a}}$	`\text{\r{a}}`	$\text{\r{a}}$
`\text{\H{o}}`	$\text{\H{o}}$				

`\overleftarrow{AB}`	$\overleftarrow{AB}$	`\overrightarrow{AB}`	$\overrightarrow{AB}$
`\underleftarrow{AB}`	$\underleftarrow{AB}$	`\underrightarrow{AB}`	$\underrightarrow{AB}$
`\overleftrightarrow{AB}`	$\overleftrightarrow{AB}$	`\underleftrightarrow{AB}`	$\underleftrightarrow{AB}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\widecheck{ac}` `\overgroup{AB}` `\undergroup{AB}`
`\overlinesegment{AB}` `\underlinesegment{AB}` `\Overrightarrow{AB}`
`\overleftharpoon{ac}` `\overrightharpoon{ac}`

Delimiters

`(~)`	$(\,)$	`\lparen~\rparen`	$(\,)$	`\lceil~\rceil`	$\lceil\,\rceil$
`\uparrow`	$\uparrow$	`[~]`	$[\,]$	`\lbrack~\rbrack`	$[\,]$
`\lfloor~\rfloor`	$\lfloor\,\rfloor$	`\downarrow`	$\downarrow$	`\{ \}`	$\{\}$
`\lbrace \rbrace`	$\{\}$	`\updownarrow`	$\updownarrow$	`\langle~\rangle`	$\langle\,\rangle$
`\lgroup~\rgroup`	$(\,)$	`\Uparrow`	$\Uparrow$	`\vert`	$\vert$
`\Downarrow`	$\Downarrow$	`\Vert`	$\Vert$	`\Updownarrow`	$\Updownarrow$
`\lvert~\rvert`	$\lvert\,\rvert$	`\lVert~\rVert`	$\lVert\,\rVert$	`\backslash`	$\backslash$
`/`	$/$				

`\lmoustache \rmoustache`	$\lmoustache\,\rmoustache$	`\ulcorner \urcorner`	$\ulcorner\,\urcorner$
`\llcorner \lrcorner`	$\llcorner\,\lrcorner$	`\llbracket~\rrbracket`	$[\![\,]\!]$

`( \big( \Big( \bigg( \Bigg(` $(\,\big(\,\Big(\,\bigg(\,\Bigg($

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\lang~\rang` `\left\lt~\right\gt` `\lBrace~\rBrace`
`\left(\LARGE{AB}\right)`

Environments

```
\begin{matrix} a & b \\ c & d \end{matrix}
```

$$\begin{matrix} a & b \\ c & d \end{matrix}$$

```
\begin{array}{cc}a & b\\c & d\end{array}
```

$$\begin{array}{cc}a & b\\c & d\end{array}$$

```
\begin{pmatrix} a & b \\ c & d \end{pmatrix}
```

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

```
\begin{bmatrix} a & b \\ c & d \end{bmatrix}
```

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

```
\begin{vmatrix} a & b \\ c & d \end{vmatrix}
```

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

```
\begin{Vmatrix} a & b \\ c & d \end{Vmatrix}
```

$$\begin{Vmatrix} a & b \\ c & d \end{Vmatrix}$$

```
\begin{Bmatrix} a & b \\ c & d \end{Bmatrix}
```

$$\begin{Bmatrix} a & b \\ c & d \end{Bmatrix}$$

```
x = \begin{cases} a &\text{if } b \\ c &\text{if } d \end{cases}
```

$$x = \begin{cases} a & \text{if } b \\ c & \text{if } d \end{cases}$$

```
\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}
```

$$\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}$$

```
\sum_{\begin{subarray}{l} i\in\Lambda\\ 0<j<n\end{subarray}}
```

$$\sum_{\begin{subarray}{l} i\in\Lambda\\ 0<j<n\end{subarray}}$$

```
\begin{CD}A @>a>> B \\@VbVV @AAcA\\C @= D\end{CD}
```

$$\begin{CD}A @>a>> B \\@VbVV @AAcA\\C @= D\end{CD}$$

```
  \begin{equation}\begin{split}a
&=b+c\\&=e+f\end{split}\end{equation}
```

$$\begin{equation}\begin{split}a &=b+c\\&=e+f\end{split}\end{equation}$$

Environments (continued)

```
\begin{align} a&=b+c \\ d+e&=f \end{align}
```

$$a = b + c \tag{1}$$
$$d + e = f \tag{2}$$

```
\begin{gather} a=b \\ e=b+c \end{gather}
```

$$a = b \tag{3}$$
$$e = b + c \tag{4}$$

```
\begin{alignat}{2}10&x+&3&y=2\\3&x+&13&y=4\end{alignat}
```

$$10x + 3y = 2 \tag{5}$$
$$3x + 13y = 4 \tag{6}$$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\def\arraystretch{1.5}\begin{array}{c:c:c} a & b & c \\ \hline d & e & f \\ \hdashline g & h & i \end{array}
```

HTML

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\url{https://katex.org/}     \htmlId{bar}{x}              \htmlClass{foo}{x}
\htmlStyle{color: red;}{x}   \htmlData{foo=a, bar=b}{x}
\href{https://katex.org/}{\KaTeX}
```

Letters and Unicode

\Gamma	Γ	\Delta	Δ	\Theta	Θ
\Lambda	Λ	\Xi	Ξ	\Pi	Π
\Sigma	Σ	\Upsilon	Υ	\Phi	Φ
\Psi	Ψ	\Omega	Ω	\varGamma	Γ
\varDelta	Δ	\varTheta	Θ	\varLambda	Λ
\varXi	Ξ	\varPi	Π	\varSigma	Σ
\varUpsilon	Υ	\varPhi	Φ	\varPsi	Ψ
\varOmega	Ω	\alpha	α	\beta	β
\gamma	γ	\delta	δ	\epsilon	ϵ
\zeta	ζ	\eta	η	\theta	θ
\iota	ι	\kappa	κ	\lambda	λ
\mu	μ	\nu	ν	\xi	ξ
\pi	π	\rho	ρ	\sigma	σ
\tau	τ	\upsilon	υ	\phi	ϕ
\chi	χ	\psi	ψ	\omega	ω
\varepsilon	ε	\varkappa	ϰ	\vartheta	ϑ
\varpi	ϖ	\varrho	ϱ	\varsigma	ς
\varphi	φ	\digamma	ϝ	\imath	ı
\nabla	∇	\Im	ℑ	\text{\OE}	Œ
\jmath	ȷ	\partial	∂	\wp	℘
\text{\o}	ø	\aleph	ℵ	\Game	⅁
\Bbbk	𝕜	\text{\O}	Ø	\Finv	Ⅎ
\text{\ss}	ß	\text{\aa}	å	\text{\i}	ı
\beth	ב	\text{\AA}	Å	\text{\j}	ȷ
\gimel	ג	\ell	ℓ	\Re	ℜ
\text{\ae}	æ	\daleth	ד	\hbar	ℏ

Letters and Unicode (continued)

```
\text{\AE} Æ   \eth ð   \hslash ħ
\text{\oe} œ
```

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\Alpha     \Beta        \Epsilon
\Zeta      \Eta         \Iota
\Kappa     \Mu          \Nu
\Omicron   \Rho         \Tau
\Chi       \omicron     \thetasym
\Reals     \image       \weierp
\alef      \N           \Z
\alefsym   \cnums       \natnums
\Complex   \R           \real
\reals     \char"263a
```

Layout

Annotation

```
\cancel{5} 5̸   \bcancel{5}  5̷   \xcancel{ABC} ABC
\not =     ≠   \text{\sout{abc}} abc
```

```
\overbrace{a+b+c}^{\text{note}}  $\overbrace{a+b+c}^{\text{note}}$   \boxed{\pi=\frac c d}  $\boxed{\pi=\frac{c}{d}}$
```

```
\text{\textcircled{a}}  ⓐ
```

```
\underbrace{a+b+c}_{\text{note}}
```

$$\underbrace{a+b+c}_{\text{note}}$$

```
\overbracket{a+b+c}^{\text{note}}
```

$$\overbracket{a+b+c}^{\text{note}}$$

```
\underbracket{a+b+c}_{\text{note}}
```

$$\underbracket{a+b+c}_{\text{note}}$$

```
\tag{hi} x+y^{2x}
```

$$x+y^{2x} \tag{hi}$$

```
\tag*{hi} x+y^{2x}
```

$$x+y^{2x} \tag*{hi}$$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\phase{30^\circ}   \angl{n}   \angln
```

Vertical Layout

`x_n` x_n `\stackrel{!}{=}` $\stackrel{!}{=}$ `a \atop b` $\begin{matrix}a\\b\end{matrix}$

`e^x` e^x `\overset{!}{=}` $\overset{!}{=}$ `_u^o` $\begin{matrix}o\\u\end{matrix}$

`\underset{!}{=}` $\underset{!}{=}$

`a\raisebox{0.25em}{$b$}c` $a^{b}c$ `\reflectbox{KaTeX}` XeTaK

`\sum_{\substack{0<i<m\\0<j<n}}` $\sum_{\substack{0<i<m\\0<j<n}}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\mathreflectbox{x^2}` `\vcenter{x}`

Overlap and Spacing

`{=}\mathllap{/\,}` $=\!\!\!/$ `\mathrlap{\,/}{=}` $=\!\!\!/$

`\left(x^{\smash{2}}\right)` $\left(x^{\smash{2}}\right)$ `\sqrt{\smash[b]{y}}` $\sqrt{\smash[b]{y}}$

`\displaystyle\sum_{\mathclap{1\le i\le j\le n}} x_{ij}`

$$\displaystyle\sum_{\mathclap{1\le i\le j\le n}} x_{ij}$$

Logic and Set Theory

`\forall`	$\forall$	`\complement`	$\complement$	`\therefore`	$\therefore$
`\emptyset`	$\emptyset$	`\exists`	$\exists$	`\subset`	$\subset$
`\because`	$\because$	`\supset`	$\supset$	`\mapsto`	$\mapsto$
`\varnothing`	$\varnothing$	`\nexists`	$\nexists$	`\mid`	$\mid$
`\to`	$\to$	`\implies`	$\implies$	`\in`	$\in$
`\land`	$\land$	`\gets`	$\gets$	`\impliedby`	$\impliedby$
`\lor`	$\lor$	`\leftrightarrow`	$\leftrightarrow$	`\iff`	$\iff$
`\notin`	$\notin$	`\ni`	$\ni$	`\neg`	$\neg$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\empty` `\exist` `\isin`
`\notni` `\set{x\VERT x<5}`
`\Set{ x \VERT x<\frac 1 2 }`

Macros

`\def\foo{x^2} \foo + \foo` $x^2 + x^2$ `\futurelet\q\mathrm x\q` xx
`\def\a{1}\xdef\b{\a 2}\b` 12 `\TextOrMath{in text}{in math}` *inmath*

`\gdef\foo#1{#1^2} \foo{y} + \foo{y}`

$y^2 + y^2$

`\def\f#1{[#1]}\def\g{abc}\expandafter\f\g`

[*a*]*bc*

`\def\a{1}\edef\b{\a 2}\def\a{9}\b`

12

`\def\a{1}\edef\b{\noexpand\a 2}\def\a{9}\b`

92

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\def\foo{\@ifstar{S}{N}}\foo*\;\foo`
`\def\foo{\@ifnextchar[{Y}{N}}\foo[x]\;\foo`
`\@firstoftwo{a}{b}\;\@secondoftwo{a}{b}`

Operators

Big Operators

`\sum`	$\sum$	`\prod`	$\prod$	`\bigotimes`	$\bigotimes$
`\bigvee`	$\bigvee$	`\int`	$\int$	`\coprod`	$\coprod$
`\bigoplus`	$\bigoplus$	`\bigwedge`	$\bigwedge$	`\iint`	$\iint$
`\intop`	$\int$	`\bigodot`	$\bigodot$	`\bigcap`	$\bigcap$
`\iiint`	$\iiint$	`\smallint`	$\int$	`\biguplus`	$\biguplus$
`\bigcup`	$\bigcup$	`\oint`	$\oint$	`\bigsqcup`	$\bigsqcup$

Drawn by the package; LaTeX has no equivalent, so no output is shown:
`\oiint` `\oiiint`

Binary Operators

`+`	$+$	`\cdot`	$\cdot$	`\gtrdot`	$\gtrdot$
`x \pmod a`	$x \pmod a$	`-`	$-$	`\cdotp`	$\cdotp$
`\intercal`	$\intercal$	`x \pod a`	$x \pod a$	`/`	$/$
`\centerdot`	$\centerdot$	`\land`	$\land$	`\rhd`	$\rhd$
`*`	$*$	`\circ`	$\circ$	`\leftthreetimes`	$\leftthreetimes$
`\rightthreetimes`	$\rightthreetimes$	`\amalg`	$\amalg$	`\circledast`	$\circledast$
`\ldotp`	$\ldotp$	`\rtimes`	$\rtimes$	`\And`	$\And$
`\circledcirc`	$\circledcirc$	`\lor`	$\lor$	`\setminus`	$\setminus$
`\ast`	$\ast$	`\circleddash`	$\circleddash$	`\lessdot`	$\lessdot$
`\smallsetminus`	$\smallsetminus$	`\barwedge`	$\barwedge$	`\Cup`	$\Cup$
`\lhd`	$\lhd$	`\sqcap`	$\sqcap$	`\bigcirc`	$\bigcirc$
`\cup`	$\cup$	`\ltimes`	$\ltimes$	`\sqcup`	$\sqcup$
`\bmod`	$\bmod$	`\curlyvee`	$\curlyvee$	`x \mod a`	$x \mod a$
`\times`	$\times$	`\boxdot`	$\boxdot$	`\curlywedge`	$\curlywedge$
`\mp`	$\mp$	`\unlhd`	$\unlhd$	`\boxminus`	$\boxminus$
`\div`	$\div$	`\odot`	$\odot$	`\unrhd`	$\unrhd$
`\boxplus`	$\boxplus$	`\divideontimes`	$\divideontimes$	`\ominus`	$\ominus$
`\uplus`	$\uplus$	`\boxtimes`	$\boxtimes$	`\dotplus`	$\dotplus$
`\oplus`	$\oplus$	`\vee`	$\vee$	`\bullet`	$\bullet$
`\doublebarwedge`	$\doublebarwedge$	`\otimes`	$\otimes$	`\veebar`	$\veebar$
`\Cap`	$\Cap$	`\doublecap`	$\Cap$	`\oslash`	$\oslash$
`\wedge`	$\wedge$	`\cap`	$\cap$	`\doublecup`	$\Cup$
`\pm`	$\pm$	`\wr`	$\wr$		

Fractions and Binomials

`\frac{a}{b}`	$\frac{a}{b}$	`\tfrac{a}{b}`	$\tfrac{a}{b}$	`{a \over b}`	$\{a \over b\}$
`\dfrac{a}{b}`	$\dfrac{a}{b}$	`a/b`	a/b	`\binom{n}{k}`	$\binom{n}{k}$
`\dbinom{n}{k}`	$\dbinom{n}{k}$	`{n\brace k}`	$\{n\brace k\}$	`{n \choose k}`	$\{n \choose k\}$
`\tbinom{n}{k}`	$\tbinom{n}{k}$	`{n\brack k}`	$\{n\brack k\}$		

`\genfrac ( ] {2pt}{1}a{a+1}` $\left(\genfrac{}{}{2pt}{1}{a}{a+1}\right]$ `\cfrac{a}{1 + \cfrac{1}{b}}` $\cfrac{a}{1+\cfrac{1}{b}}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:
`{a \above{2pt} b+1}`

Math Operators

`\arcsin`	$\arcsin$	`\deg`	$\deg$	`\sec`	$\sec$
`\arccos`	$\arccos$	`\cosh`	$\cosh$	`\dim`	$\dim$
`\sin`	$\sin$	`\arctan`	$\arctan$	`\cot`	$\cot$
`\exp`	$\exp$	`\sinh`	$\sinh$	`\hom`	$\hom$
`\coth`	$\coth$	`\ker`	$\ker$	`\tan`	$\tan$
`\arg`	$\arg$	`\csc`	$\csc$	`\lg`	$\lg$
`\tanh`	$\tanh$	`\ln`	$\ln$	`\cos`	$\cos$
`\log`	$\log$	`\operatorname{f}`	f	`\injlim`	$\injlim$
`\min`	$\min$	`\varinjlim`	$\varinjlim$	`\lim`	$\lim$
`\varliminf`	$\varliminf$	`\det`	$\det$	`\liminf`	$\liminf$
`\Pr`	$\Pr$	`\varlimsup`	$\varlimsup$	`\gcd`	$\gcd$
`\limsup`	$\limsup$	`\projlim`	$\projlim$	`\varprojlim`	$\varprojlim$
`\inf`	$\inf$	`\max`	$\max$	`\sup`	$\sup$

`\operatorname*{f}` $\operatorname*{f}$

`\operatornamewithlimits{f}` $\operatornamewithlimits{f}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\cosec` `\arctg` `\cotg`
`\sh` `\arcctg` `\ch`
`\ctg` `\tg` `\cth`
`\th` `\argmax` `\argmin`
`\plim`

Square roots

`\sqrt{x}` $\sqrt{x}$ `\sqrt[3]{x}` $\sqrt[3]{x}$

Relations

`\stackrel{!}{=}`	$\stackrel{!}{=}$	`=`	$=$	`\doteqdot`	$\doteqdot$
`\lessapprox`	$\lessapprox$	`\smile`	$\smile$	`<`	$<$
`\eqcirc`	$\eqcirc$	`\lesseqgtr`	$\lesseqgtr$	`\sqsubset`	$\sqsubset$
`>`	$>$	`\eqcolon`	$\eqcolon$	`\lesseqqgtr`	$\lesseqqgtr$
`\sqsubseteq`	$\sqsubseteq$	`:`	$:$	`\Eqcolon`	$\Eqcolon$
`\lessgtr`	$\lessgtr$	`\sqsupset`	$\sqsupset$	`\approx`	$\approx$
`\eqqcolon`	$\eqqcolon$	`\lesssim`	$\lesssim$	`\sqsupseteq`	$\sqsupseteq$
`\approxcolon`	$\approxcolon$	`\Eqqcolon`	$\Eqqcolon$	`\ll`	$\ll$
`\Subset`	$\Subset$	`\eqsim`	$\eqsim$	`\lll`	$\lll$
`\subset`	$\subset$	`\approxeq`	$\approxeq$	`\eqslantgtr`	$\eqslantgtr$
`\llless`	$\llless$	`\subseteq`	$\subseteq$	`\asymp`	$\asymp$
`\eqslantless`	$\eqslantless$	`\subseteqq`	$\subseteqq$	`\backepsilon`	$\backepsilon$
`\equiv`	$\equiv$	`\mid`	$\mid$	`\succ`	$\succ$
`\backsim`	$\backsim$	`\fallingdotseq`	$\fallingdotseq$	`\models`	$\models$
`\succapprox`	$\succapprox$	`\backsimeq`	$\backsimeq$	`\frown`	$\frown$
`\multimap`	$\multimap$	`\succcurlyeq`	$\succcurlyeq$	`\between`	$\between$
`\ge`	$\ge$	`\succeq`	$\succeq$	`\bowtie`	$\bowtie$
`\geq`	$\geq$	`\owns`	$\owns$	`\succsim`	$\succsim$
`\bumpeq`	$\bumpeq$	`\geqq`	$\geqq$	`\parallel`	$\parallel$
`\Supset`	$\Supset$	`\Bumpeq`	$\Bumpeq$	`\geqslant`	$\geqslant$
`\perp`	$\perp$	`\supset`	$\supset$	`\circeq`	$\circeq$
`\gg`	$\gg$	`\pitchfork`	$\pitchfork$	`\supseteq`	$\supseteq$
`\colonapprox`	$\colonapprox$	`\ggg`	$\ggg$	`\prec`	$\prec$
`\supseteqq`	$\supseteqq$	`\Colonapprox`	$\Colonapprox$	`\gggtr`	$\gggtr$

Relations (continued)

\precapprox	⪷	\thickapprox	≈	\coloneq	≔
\preccurlyeq	≼	\thicksim	∼	\Coloneq	⩴
\gtrapprox	⪆	\preceq	⪯	\trianglelefteq	⊴
\coloneqq	≔	\gtreqless	⋚	\precsim	≾
\triangleq	≜	\Coloneqq	⩴	\gtreqqless	⪌
\propto	∝	\trianglerighteq	⊵	\colonsim	∷∼
\gtrless	≷	\risingdotseq	≓	\varpropto	∝
\Colonsim	∷∼	\gtrsim	≳	\shortmid	∣
\vartriangle	△	\cong	≅	\shortparallel	∥
\vartriangleleft	◁	\curlyeqprec	⋞	\in	∈
\sim	∼	\vartriangleright	▷	\curlyeqsucc	⋟
\Join	⋈	\simcolon	∼:	\vcentcolon	:
\dashv	⊣	\le	≤	\vdash	⊢
\dblcolon	::	\leq	≤	\simeq	≃
\vDash	⊨	\doteq	≐	\leqq	≦
\smallfrown	⌢	\Vdash	⊩	\Doteq	≑
\leqslant	⩽	\smallsmile	⌣	\Vvdash	⊪

Drawn by the package; LaTeX has no equivalent, so no output is shown:

\approxcoloncolon \lt \origof
\gt \imageof \simcoloncolon

Negated Relations

\not =	≠	\gnapprox	⪊	\ngeqslant	⩾̸
\nsubseteq	⊈	\precneqq	⪵	\gneq	⪈
\ngtr	≯	\nsubseteqq	⫅̸	\precnsim	⋨
\gneqq	≩	\nleq	≰	\nsucc	⊁
\subsetneq	⊊	\gnsim	⋧	\nleqq	≦̸
\nsucceq	⋡	\subsetneqq	⫋	\gvertneqq	≩
\nleqslant	⩽̸	\nsupseteq	⊉	\succnapprox	⪺
\lnapprox	⪉	\nless	≮	\nsupseteqq	⫆̸
\succneqq	⪶	\lneq	⪇	\nmid	∤
\ntriangleleft	⋪	\succnsim	⋩	\lneqq	≨
\notin	∉	\ntrianglelefteq	⋬	\supsetneq	⊋
\lnsim	⋦	\ntriangleright	⋫	\supsetneqq	⫌
\lvertneqq	≨	\nparallel	∦	\ntrianglerighteq	⋭
\varsubsetneq	⊊	\ncong	≇	\nprec	⊀
\nvdash	⊬	\varsubsetneqq	⫋	\ne	≠
\npreceq	⋠	\nvDash	⊭	\varsupsetneq	⊋
\neq	≠	\nshortmid	∤	\nVDash	⊯
\varsupsetneqq	⫌	\ngeq	≱	\nshortparallel	∦
\nVdash	⊮	\ngeqq	≧̸	\nsim	≁
\precnapprox	⪹				

Drawn by the package; LaTeX has no equivalent, so no output is shown:

\notni

Arrows

\circlearrowleft	↺	\leftharpoonup	↼	\circlearrowright	↻
\leftleftarrows	⇇	\curvearrowleft	↶	\leftrightarrow	↔
\restriction	↾	\curvearrowright	↷	\Leftrightarrow	⇔
\rightarrow	→	\leftrightarrows	⇆	\Rightarrow	⇒
\rightarrowtail	↣	\rightharpoondown	⇁	\dashleftarrow	⇠
\Lleftarrow	⇚	\rightharpoonup	⇀	\dashrightarrow	⇢
\longleftarrow	⟵	\rightleftarrows	⇄	\downarrow	↓
\Longleftarrow	⟸	\Downarrow	⇓	\rightrightarrows	⇉
\downdownarrows	⇊	\rightsquigarrow	⇝	\downharpoonleft	⇃
\longmapsto	⟼	\Rrightarrow	⇛	\downharpoonright	⇂
\longrightarrow	⟶	\Rsh	↱	\gets	←
\Longrightarrow	⟹	\searrow	↘	\looparrowleft	↫
\swarrow	↙	\looparrowright	↬	\to	→
\twoheadleftarrow	↞	\hookleftarrow	↩	\hookrightarrow	↪
\iff	⟺	\Lsh	↰	\impliedby	⟸
\mapsto	↦	\implies	⟹	\nearrow	↗
\uparrow	↑	\nleftarrow	↚	\Uparrow	⇑
\nLeftarrow	⇍	\updownarrow	↕	\nleftrightarrow	↮
\Updownarrow	⇕	\leadsto	⇝	\nLeftrightarrow	⇎
\upharpoonleft	↿	\leftarrow	←	\nrightarrow	↛
\upharpoonright	↾	\Leftarrow	⇐	\nRightarrow	⇏
\upuparrows	⇈	\leftarrowtail	↢	\nwarrow	↖
\mapsfrom	↤	\leftharpoondown	↽	\xleftarrow{abc}	←abc
\xLeftarrow{abc}	⟸abc	\xRightarrow{abc}	⟹abc	\xmapsto{abc}	⟼abc

\leftrightharpoons	⇋	\leftrightsquigarrow	↭
\rightleftharpoons	⇌	\longleftrightarrow	⟷
\Longleftrightarrow	⟺	\twoheadrightarrow	↠
\xrightarrow[under]{over}	over / under	\xleftrightarrow{abc}	⟷abc
\xLeftrightarrow{abc}	⟺abc	\xhookleftarrow{abc}	↩abc
\xhookrightarrow{abc}	↪abc	\xleftharpoonup{abc}	↼abc
\xrightharpoonup{abc}	⇀abc	\xleftharpoondown{abc}	↽abc
\xrightharpoondown{abc}	⇁abc	\xleftrightharpoons{abc}	⇋abc
\xrightleftharpoons{abc}	⇌abc		

Drawn by the package; LaTeX has no equivalent, so no output is shown:

\rArr	\rarr	\Darr
\dArr	\darr	\Harr
\hArr	\harr	\Lrarr
\lrArr	\lrarr	\Uarr
\uArr	\uarr	\Larr
\lArr	\larr	\Rarr
\xlongequal{abc}	\xtofrom{abc}	\xtwoheadrightarrow{abc}
\xtwoheadleftarrow{abc}		

Special Notation

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\bra{\phi}   \ket{\psi}   \braket{\phi\VERT\psi}
\Bra{\phi}   \Ket{\psi}
```

Style, Color, Size, and Font

`\color{blue} F=ma`	$F = ma$	`\mathrm{Ab0}`	Ab0	`\mathbf{Ab0}`	**Ab0**
`\mathsf{Ab0}`	Ab0	`\mathnormal{Ab0}`	*Ab0*	`\textbf{Ab0}`	**Ab0**
`\textsf{Ab0}`	Ab0	`\textrm{Ab0}`	Ab0	`\bf Ab0`	**Ab0**
`\sf Ab0`	Ab0	`\rm Ab0`	Ab0	`\bold{Ab0}`	**Ab0**
`\textnormal{Ab0}`	Ab0	`\boldsymbol{Ab0}`	**Ab0**	`\Bbb{AB}`	$\mathbb{AB}$
`\text{Ab0}`	Ab0	`\bm{Ab0}`	**Ab0**	`\mathbb{AB}`	$\mathbb{AB}$
`\textup{Ab0}`	Ab0	`\textmd{Ab0}`	Ab0	`\frak{Ab0}`	$\mathfrak{Ab0}$
`\mathit{Ab0}`	*Ab0*	`\mathtt{Ab0}`	Ab0	`\mathfrak{Ab0}`	$\mathfrak{Ab0}$
`\textit{Ab0}`	*Ab0*	`\texttt{Ab0}`	Ab0	`\mathcal{AB0}`	$\mathcal{AB0}$
`\it Ab0`	*Ab0*	`\tt Ab0`	Ab0	`\cal AB0`	$\mathcal{AB0}$
`\emph{Ab0}`	*Ab0*	`\mathscr{AB}`	$\mathscr{AB}$	`\scriptstyle x`	x
`\lim\limits_x`	$\lim_x$	`\lim\nolimits_x`	$\lim_x$	`\verb!x^2!`	x^2

`\textcolor{blue}{F=ma}`	$F = ma$	`\textsf{\textbf{H}}`	**H**
`\mathsf{\mathbf{H}}`	**H**	`\displaystyle\sum_{i=1}^n`	$\displaystyle\sum_{i=1}^{n}$
`\textstyle\sum_{i=1}^n`	$\textstyle\sum_{i=1}^{n}$	`\scriptscriptstyle x`	x
`\mathchoice{D}{T}{S}{SS}`	T		

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\textcolor{#228B22}{F=ma}   \colorbox{aqua}{$F=ma$}   \mathsfit{Ab0}
\pmb{\mu}                   \Huge AB                   \normalsize AB
\huge AB                    \small AB                  \LARGE AB
\footnotesize AB            \Large AB                  \scriptsize AB
\large AB                   \tiny AB                   \sixptsize sixptsize
\fcolorbox{red}{aqua}{$F=ma$}
```

Symbols and Punctuation

`\dots`	…	`\%`	%	`\cdots`	⋯
`\#`	#	`\ddots`	⋱	`\&`	&
`\ldots`	…	`\nabla`	∇	`\_`	_
`\vdots`	⋮	`\infty`	∞	`\dotsb`	⋯
`\text{--}`	–	`\dotsc`	…	`\checkmark`	✓
`\dotsi`	⋯	`\dag`	†	`\text{---}`	—
`\dotsm`	⋯	`\dagger`	†	`\dotso`	…
`\ddag`	‡	`\mathellipsis`	…	`\ddagger`	‡
`` ` ``	`	`\Box`	□	`\lq`	‘
`\square`	□	`\angle`	∠	`\blacksquare`	■
`\measuredangle`	∡	`\rq`	’	`\triangle`	△
`\sphericalangle`	∢	`\triangledown`	▽	`\top`	⊤
`"`	"	`\triangleleft`	◁	`\bot`	⊥
`\triangleright`	▷	`\$`	$	`\colon`	:
`\bigtriangledown`	▽	`\backprime`	‵	`\bigtriangleup`	△
`\pounds`	£	`\prime`	′	`\blacktriangle`	▲
`\mathsterling`	£	`\text{\textless}`	<	`\yen`	¥
`\text{\textbar}`	|	`\surd`	√	`\diamond`	⋄
`\Diamond`	◇	`\lozenge`	◊	`\mho`	℧

Symbols and Punctuation (continued)

\blacklozenge	⧫	\diagdown	╲	\text{\P}	¶
\star	★	\diagup	╱	\text{\S}	§
\bigstar	★	\flat	♭	\clubsuit	♣
\natural	♮	\copyright	©	\sharp	♯
\circledR	®	\diamondsuit	♢	\heartsuit	♡
\circledS	Ⓢ	\spadesuit	♠	\maltese	✠
\minuso	⦵				

\text{\textunderscore}	_	\text{\textendash}	–
\text{\textemdash}	—	\text{\textdagger}	†
\text{\textasciitilde}	~	\text{\textasciicircum}	^
\text{\textellipsis}	…	\text{\textdaggerdbl}	‡
\text{\textquoteleft}	‘	\text{\textquoteright}	’
\text{\textquotedblleft}	“	\text{\textquotedblright}	”
\text{\textdollar}	$	\blacktriangledown	▼
\text{\textsterling}	£	\text{\textgreater}	>
\blacktriangleleft	◀	\blacktriangleright	▶
\text{\textbardbl}	‖	\text{\textbraceleft}	{
\text{\textdegree}	°	\text{\textbraceright}	}
\text{\textbackslash}	\	\text{\textregistered}	®

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\KaTeX      \LaTeX      \TeX
\infin      \sdot       \Dagger
\degree     \text{\sect} \clubs
\diamonds   \hearts     \spades
```

Units

\rule{1em}{1em} ■ \rule{10pt}{10pt} ■

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\huge\rule{1em}{1em}     \rule{18mu}{18mu}   \huge\rule{18mu}{18mu}
\huge\rule{10pt}{10pt}
\scriptscriptstyle\rule{1em}{1em}
\scriptscriptstyle\rule{18mu}{18mu}
\scriptscriptstyle\rule{10pt}{10pt}
```

Extensions

mhchem

\ce{H2O} $\mathrm{H_2O}$ \ce{1/2 H2O} $\frac{1}{2}\,\mathrm{H_2O}$ \ce{CuSO4*5H2O} $\mathrm{CuSO_4 \cdot 5\,H_2O}$

\ce{CO2 + C -> 2 CO} $\mathrm{CO_2 + C \longrightarrow 2\,CO}$ \ce{^{227}_{90}Th+} $\mathrm{^{227}_{90}Th^{+}}$

\ce{Fe^{3+} + 3e- -> Fe(s)} $\mathrm{Fe^{3+} + 3\,e^{-} \longrightarrow Fe(s)}$

\ce{SO4^2- + Ba^2+ -> BaSO4 v}

$\mathrm{SO_4^{2-} + Ba^{2+} \longrightarrow BaSO_4\downarrow}$

\ce{Zn^2+ <=>[+ 2OH-][+ 2H+] Zn(OH)2 v}

$\mathrm{Zn^{2+} \underset{+2\,H^{+}}{\overset{+2\,OH^{-}}{\rightleftharpoons}} Zn(OH)_2\downarrow}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\pu{1.5 mol/L}   \pu{123 kJ/mol}   \pu{1.2e-3 mol L-1}
```

Further LaTeX

More commands

`\idotsint` $\int\cdots\int$ `\iiiint` $\iiint\!\!\int$

```
\text{\textsection\textparagraph\textcopyright\texttrademark}
```

§¶©™

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\euro
```

Numbers and units (a subset of siunitx)

`\num{12345.6789}` 12 345.6789 `\num{1.5e-4}` 1.5×10^{-4} `\num{1.2(3)e3}` $1.2(3)\times10^{3}$

`\si{kg.m.s^-2}` $\mathrm{kg\,m\,s^{-}\,2}$ `\ang{1;2;3}` $1^{\circ}2^{\prime}3^{\prime\prime}$ `\numrange{1}{10}` 1 to 10

`\numlist{1;2;3}` 1, 2, and 3

```
\qty{1.5}{\kilo\gram}
```

1.5 kg `\SIrange{1}{5}{\volt}` 1 V to 5 V

```
\SI{9.81}{\meter\per\second\squared}
```

$9.81\,\mathrm{m\,s^{-2}}$

Equations

`\cancelto{0}{x}` $\cancelto{0}{x}$

```
\bigl\lvert x \bigr\rvert
```

$\bigl\lvert x \bigr\rvert$

```
\begin{dcases} x & x>0 \\ -x & \text{otherwise} \end{dcases}
```

$$\begin{cases} x & x>0 \\ -x & \text{otherwise} \end{cases}$$

```
\begin{subarray}{c} i<n \\ j<m \end{subarray}
```

$\begin{subarray}{c} i<n \\ j<m \end{subarray}$

```
\sqrt[\leftroot{-2}\uproot{2}n]{x}
```

$\sqrt[\leftroot{-2}\uproot{2}n]{x}$

```
a = b \tag{1}
```

$$a = b \tag{1}$$

```
a = b \tag*{(*)}
```

$$a = b \tag*{(*)}$$

Tables

```
\begin{tabular}{|l|r|} \hline a & 1 \\ \hline b & 22 \\ \hline
\end{tabular}
```

a	1
b	22

```
\begin{array}{l} \textbf{Bold} \\ \textsc{Small caps} \\
\verb|verbatim| \end{array}
```

Bold
SMALL CAPS
`verbatim`

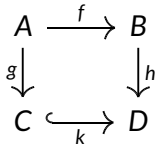
Further LaTeX / Tables (continued)

Drawn by the package; LaTeX has no equivalent, so no output is shown:

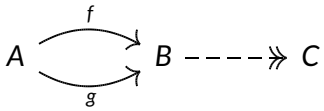
```
\begin{array}{c:c} a & b \\ \hdashline c & d \end{array}
```

Commutative diagrams

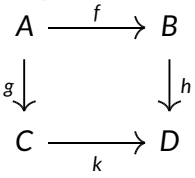
```
\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "g"'] & B \arrow[d,
"h"] \\ C \arrow[r, hook, "k"'] & D \end{tikzcd}
```




```
\begin{tikzcd}[column sep=large] A \arrow[r, "f", bend left]
\arrow[r, "g"', bend right] & B \arrow[r, two heads, dashed] & C
\end{tikzcd}
```

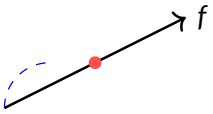



```
\begin{CD} A @>f>> B \\ @VgVV @VVhV \\ C @>>k> D \end{CD}
```

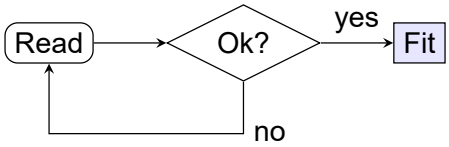


TikZ pictures

```
\begin{tikzpicture}\draw[->, thick] (0,0) -- (2,1) node[right]
{$f$}; \fill[red!70] (1,0.5) circle (2pt); \draw[blue, dashed]
(0,0) arc (180:90:0.5);\end{tikzpicture}
```




```
\begin{tikzpicture}[node distance=8mm, >=stealth] \node[draw,
rounded corners] (a) {Read}; \node[draw, diamond, aspect=2,
right=of a] (b) {Ok?}; \node[draw, fill=blue!10, right=of b] (c)
{Fit}; \draw[->] (a) -- (b); \draw[->] (b) -- node[above] {yes}
(c); \draw[->] (b) |- ++(0,-1) node[right] {no} -|
(a);\end{tikzpicture}
```

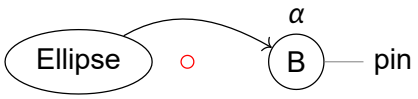



```
\begin{tikzpicture}\foreach \x/\c in
{0/red,1/orange,2/green,3/blue} {\fill[\c] (\x*0.5,0) circle
(0.2);}\end{tikzpicture}
```

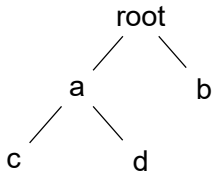


Further LaTeX / TikZ pictures (continued)

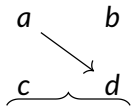
```
\begin{tikzpicture}\node[draw, ellipse] (a) at (0,0) {Ellipse};
\node[draw, circle, label=above:{$\alpha$}, pin=right:pin] (b)
at (2.4,0) {B}; \draw[->] (a) to[bend left] (b); \draw[red]
($(a)!0.5!(b)$) circle (2pt);\end{tikzpicture}
```



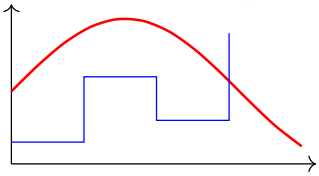

```
\begin{tikzpicture}[level distance=8mm, sibling distance=14mm]
\node {root} child {node {a} child {node {c}} child {node {d}}}
child {node {b}};\end{tikzpicture}
```



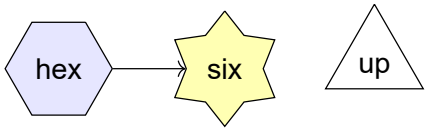

```
\begin{tikzpicture}\matrix (m) [matrix of nodes, row sep=3mm,
column sep=6mm] {$a$ & $b$ \\ $c$ & $d$ \\}; \draw[->] (m-1-1)
-- (m-2-2); \draw[decorate, decoration={brace, amplitude=4pt}]
(m-2-1.south west) -- (m-2-2.south east);\end{tikzpicture}
```



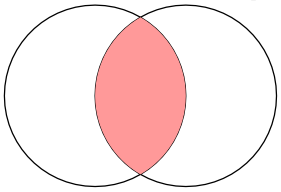

```
\begin{tikzpicture}[scale=0.8]\draw[->] (0,0) -- (4.2,0);
\draw[->] (0,0) -- (0,2.2); \draw[domain=0:4, smooth,
samples=12, red, thick] plot (\x, {1+sin(\x r)}); \draw[blue]
plot[const plot] coordinates {(0,0.3) (1,1.2) (2,0.6)
(3,1.8)};\end{tikzpicture}
```



```
\begin{tikzpicture}\node[regular polygon, regular polygon
sides=6, draw, fill=blue!10] (h) {hex}; \node[star, star
points=6, draw, fill=yellow!30, right=8mm of h] (s) {six};
\node[isosceles triangle, isosceles triangle apex angle=60,
shape border rotate=90, draw, right=8mm of s] (t) {up};
\draw[->] (h.east) -- (s.west);\end{tikzpicture}
```




```
\begin{tikzpicture}\draw (0,0) circle (1); \draw (1,0) circle
(1); \begin{scope}\clip (0,0) circle (1); \fill[red!40] (1,0)
circle (1);\end{scope}\end{tikzpicture}
```



Boxes and columns

```
\begin{tcolorbox}[title=A box] Text in a box, which is as wide
as the line. \end{tcolorbox}
```

A box

Text in a box, which is as wide as the line.

```
   \begin{tcolorbox}[colback=yellow!15, colframe=blue!50!black,
colbacktitle=blue!15, coltitle=black, title=Colours] A title
band, a frame, and a body in colours of its own. \end{tcolorbox}
```

Colours

A title band, a frame, and a body in colours of its own.

```
   \begin{multicols}{2}[A header over both columns] One paragraph
of some words that run on to a second line.\par Another, in the
first column too.\par A third, which is balanced into the
second.\par A fourth. \end{multicols}
```

A header over both columns

One paragraph of some words that run on to a second line.
Another, in the first column too.

A third, which is balanced into the second.
A fourth.